

Workout Tracker Breakdown

MVC Design

Controllers (app/controllers)

- `application_controller.rb`
 - All controllers inherit from this file
 - Has user authentication
- `workouts_controller.rb`
 - Contains 7 actions:
 - Index
 - Asks the model for all workouts to display
 - Show
 - Asks model for specific workout and its details
 - `new/edit`
 - Sets up empty objects to hand to html
 - `create/update`
 - Takes submitted form data and saves the nested record
 - Destroy
 - Deletes record from database
 - `Require_admin`
 - Method that stops guests from using crud functions
 - `Workout_params`
 - Explicitly lists the fields allowed to be saved to database
- `Workout_types_controller.rb`
 - Handles the CRUD actions for WorkoutType model
- `Exercises_controller.rb`
 - Handles the CRUD actions for the exercise model

Models

These represent our database tables and the rules that outline how the data relates to each other.

- `Application_record.rb`
 - All models inherit from this file
 - Inherits from ActiveRecord::base
 - Rails engine that translate ruby method into sql
- `User.rb`
 - The owner of data

- User has_many workouts
 - Since a user has many workouts, each database workout will have a user_id
- Workout.rb
 - The data for a single gym session
 - Belongs_to user
 - Belongs_to workout_type
 - Has_many exercise_logs
 - Has accepts nested attributes for to allow for atomic transactions regarding workout data like reps and sets
- Exercise_log.rb
 - Join table
 - This model represents the specific instance of an exercise performed and a specific day
 - Belongs to workout
 - Belongs to exercise
 - Has many workout_sets
 - A single workout has many exercises and an exercise can belong to many workouts
 - Many to many relationship
- workout_Set.rb
 - Fact table
 - Stores metrics for performed exercise
- Workout_type.rb
 - Categorizes each workout record
- exercise.rb
 - Dimension table for exercises

View

- layouts/application.html.erb
 - The layout skeleton for the entire app
 - Global nav bar
- workouts/index.html.erb
 - Dashboard for all workout sessions
 - Receives a @workout variable that contains all workouts in the database and it loops through to display them
 - Action buttons only visible to admin user
- workouts/show.html.erb
 - View displays granular data for individual workout session
 - Displays workout type designation

- Loops through exercise logs to display exercises along with the workout sets data
 - Media viewer, requests and displays image from Cloudinary database
- workouts/new.html.erb & workouts/edit.html.erb
 - Wrapper files
 - Pages for creating or modifying workouts
 - Share partial file forms to handle the same input fields
- workouts/_form.html.erb
 - Form to capture user input
 - Partial file

Authentication and Authorization

This application features a very basic security system. The site can be accessed by 3 roles of users. Admin, guest and authenticated user. Admin is the only role with full CRUD capabilities throughout the app. Guest user is a secured account that only has viewing privileges and unauthenticated user can access the page but view none of the content until authenticated as admin or guest.

- Devise
 - Encryption for passwords
 - Handles session cookies
- Role Based Access
 - Uses admin boolean on users table
 - Front end action buttons are wrapped in user admin booleans
 - Require admin methods in controllers check users whenever a page loads